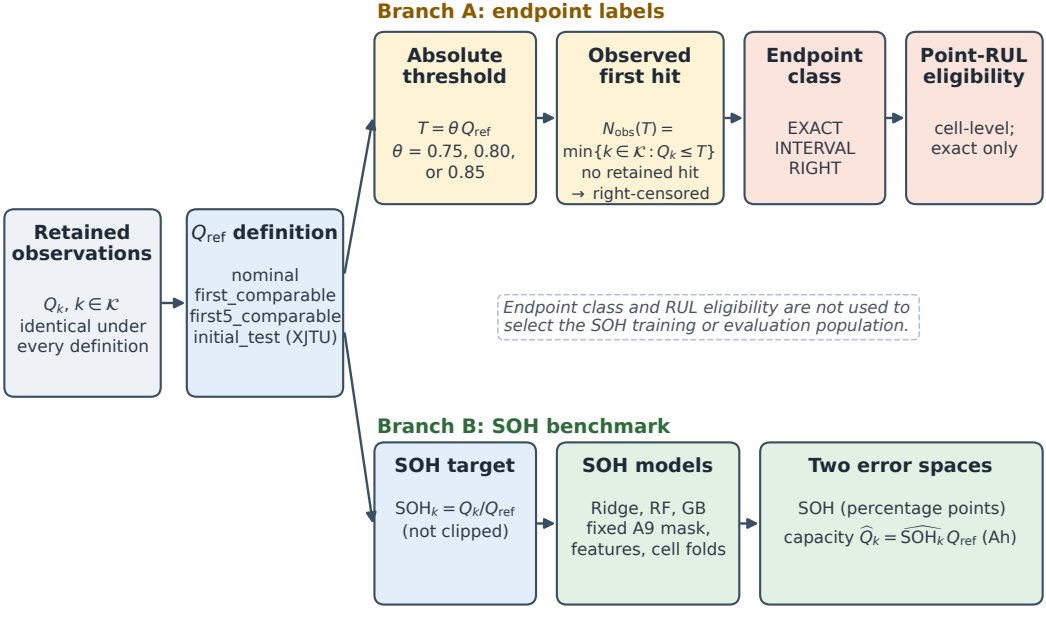
Branch A: endpoint labels
Retained observations
Q_k, $k \in \mathcal{K}$
identical under every definition
Q_{ref} definition
nominal
first_comparable
first5_comparable
initial_test (XJTU)
Absolute threshold
$T = \theta Q_{\text{ref}}$
θ = 0.75, 0.80, or 0.85
Observed first hit
$N_{\text{obs}}(T) = \min\{k \in \mathcal{K} : Q_k \le T\}$
no retained hit → right-censored
Endpoint class
EXACT
INTERVAL
RIGHT
Point-RUL eligibility
cell-level; exact only
Endpoint class and RUL eligibility are not used to select the SOH training or evaluation population.
Branch B: SOH benchmark
SOH target
$\text{SOH}_k = Q_k / Q_{\text{ref}}$
(not clipped)
SOH models
Ridge, RF, GB
fixed A9 mask, features, cell folds
Two error spaces
SOH (percentage points)
capacity $\widehat{Q}_k = \widehat{\text{SOH}}_k Q_{\text{ref}}$ (Ah)